The specific heat capacity of water is very high. One gram of water requires 1 calorie of heat to raise its temperature by 1°C. Due to its high specific heat capacity, water takes time to become hot as well as to cool down. Thus, water can absorb a lot of heat and retain it for a longer time. This property of water is used to cool engines. Water is circulated around car engine using the radiator pump and the heat is absorbed. Thus the engine is protected from getting too hot.



Figure 13.4 Water as coolant in car engines

13.2.2 Chemical properties

a. Action towards litmus paper

Pure water is neutral and it shows no action towards litmus paper.

b. Stability

Water is a very stable compound. It does not decompose into elements, when heated to ordinary temperatures. However, if it is heated to 200°C, 0.02% of water decomposes to form hydrogen and oxygen gas.

$$2H_2O \xrightarrow{2000^\circ C} 2H_2 + O_2$$

c. Catalytic nature

Water acts as a catalyst in a number of reactions. Perfectly dry hydrogen and chlorine gases do not react in the presence of sunlight. However in the presence of traces of water, the reaction takes place with explosion to produce hydrogen chloride.

$$H_2 + Cl_2 \xrightarrow[\text{Sunlight}]{\text{Moisture}} 2HCl$$

d. Reaction with metals

Water reacts with some metals. Metals such as sodium, potassium and calcium react vigorously with water at room temperature. Sodium reacts with water to form hydrogen gas and sodium hydroxide solution. Due to the heat evolved in this reaction the hydrogen gas catches fire and burns.

$$2Na + 2H_2O \longrightarrow 2NaOH + H_2$$

Activity 2

Fill a trough with water. Cut a small piece of sodium with a knife and carefully drop it in the water. Sodium reacts with water and darts across the surface of water. A flame produced is also seen near the surface.

Magnesium is little more sluggish. It reacts with hot water and gives hydrogen and magnesium hydroxide solution.

$$Mg + 2H_2O \longrightarrow Mg(OH)_2 + H_2\uparrow$$

Many other metals react with water to form oxides and hydroxides. Iron is one such metal which forms iron oxide, called rust. Iron is used in many buildings, factories, bridges, ships and vehicles. The slow and gradual rusting of iron is called corrosion.

DO YOU KNOW?

Copper does not react with water at any temperature. That is why it is used for making pipes and boilers.

e. Reaction with non-metals

Red hot carbon (coke) reacts with steam to produce water gas (Carbon monoxide + H_2).

$$C + H_2O \xrightarrow{1000^\circ C} CO\uparrow + H_2\uparrow$$

Chlorine gas dissolves in water and produces hydrochloric acid.

$$2Cl_2 + 2H_2O \xrightarrow{\text{Sunlight}} 4HCl + O_2\uparrow$$

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